

Capacitor Charging and a Power On Reset for an Active Low Microcontroller

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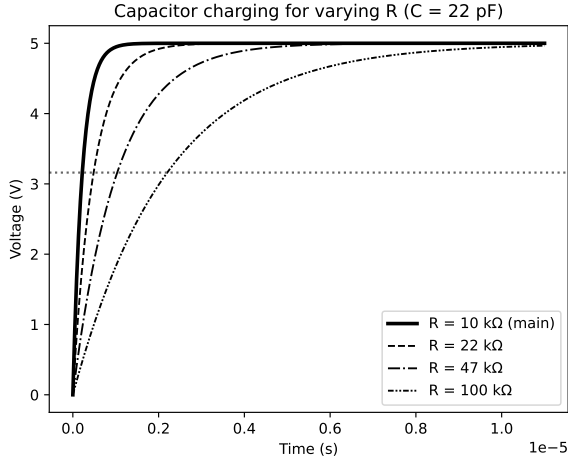


Fig. 1. Capacitor charging for different resistance values.

Abstract—This document introduces the charging behaviour of a resistor-capacitor (RC) network and applies the same charging behaviour to a power-on-reset circuit for an active-low microcontroller.

I. INTRODUCTION

When a capacitor C is charged through a series resistor R from a constant supply V_0 , the capacitor voltage rises according to

$$V(t) = V_0 \left(1 - e^{-t/RC}\right) \quad (1)$$

The product RC is called the time constant,

$$\tau = RC. \quad (2)$$

After one time constant, the capacitor reaches approximately 63.2% of the supply voltage.

II. VOLTAGE AT THE TIME CONSTANT

Starting from (1) and substituting $t = \tau$,

$$V(\tau) = V_0 \left(1 - e^{-\tau/RC}\right) \quad (3)$$

$$= V_0 \left(1 - e^{-RC/RC}\right) \quad (4)$$

$$= V_0 (1 - e^{-1}) \quad (5)$$

$$\approx V_0 (1 - 0.368) \quad (6)$$

$$\approx 0.632V_0. \quad (7)$$

For $V_0 = 5.0$ V,

$$V(\tau) \approx 0.632(5.0) \approx 3.16 \text{ V}. \quad (8)$$

III. POWER ON RESET WITH AN ACTIVE LOW RESET

A power-on-reset circuit holds the microcontroller in reset while the capacitor charges. The active-low reset pin remains low while the capacitor voltage is below the input-high threshold V_{IH} . Once the voltage crosses V_{IH} , the reset is released and the microcontroller can begin operating.

The reset pin releases when the capacitor voltage reaches V_{IH} :

$$V_{IH} = V_0 \left(1 - e^{-t_{\text{release}}/RC}\right). \quad (9)$$

Dividing by V_0 ,

$$\frac{V_{IH}}{V_0} = 1 - e^{-t_{\text{release}}/RC}. \quad (10)$$

Rearranging,

$$e^{-t_{\text{release}}/RC} = 1 - \frac{V_{IH}}{V_0}. \quad (11)$$

Taking the natural logarithm,

$$-\frac{t_{\text{release}}}{RC} = \ln \left(1 - \frac{V_{IH}}{V_0}\right). \quad (12)$$

Therefore,

$$t_{\text{release}} = -RC \ln \left(1 - \frac{V_{IH}}{V_0}\right). \quad (13)$$

Solving for R gives

$$R = \frac{-t_{\text{release}}}{C \ln \left(1 - \frac{V_{IH}}{V_0}\right)}. \quad (14)$$

For $V_{IH} = 0.7V_0$, $t_{\text{release}} = 10$ ms, and $C = 100$ nF,

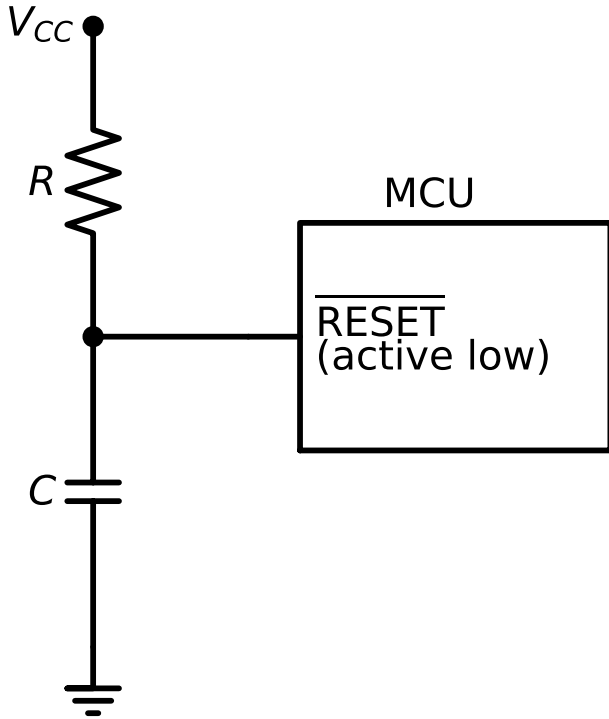


Fig. 2. RC power-on-reset circuit for an active-low reset pin.

$$\ln(1 - 0.7) = \ln(0.3) \approx -1.204. \quad (15)$$

Using (14),

$$R = \frac{-(10 \times 10^{-3})}{(100 \times 10^{-9})(-1.204)} \approx 83 \text{ k}\Omega. \quad (16)$$

An 83 k Ω resistor with a 100 nF capacitor gives a reset hold-off of approximately 10 ms.

IV. APPLICATIONS AND LIMITATIONS

A. Application

A low-cost temperature sensor node is a suitable application for an RC-only power-on-reset circuit. It only needs a short delay when power is first applied and does not require precise reset timing, brown-out detection, or re-triggering.

For this application,

$$R = 10 \text{ k}\Omega, \quad C = 220 \text{ nF}. \quad (17)$$

Assuming $V_{IH} = 0.7V_0$ and using (13),

$$t_{\text{release}} = -(10 \times 10^3)(220 \times 10^{-9}) \ln(0.3) \approx 2.65 \text{ ms}. \quad (18)$$

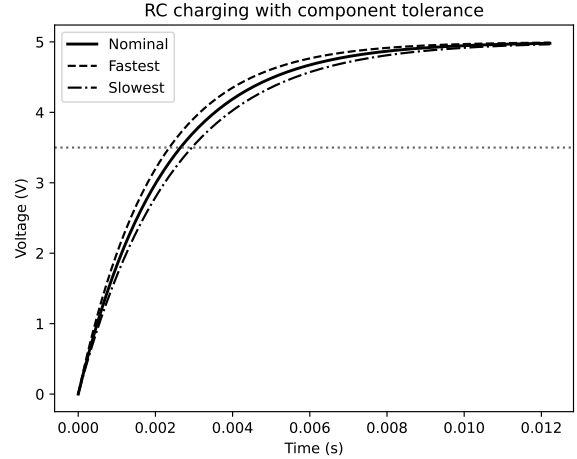


Fig. 3. Nominal, fastest, and slowest RC charging curves.

B. Limitations

One limitation is component tolerance. The actual resistor and capacitor values can vary from their nominal values, which changes the reset timing.

A second limitation is a slowly rising supply voltage. If V_{CC} rises slowly, the reset pin can remain near its threshold and may not produce a clean reset.

A third limitation is that the RC circuit does not provide brown-out detection. If the supply voltage drops after startup, the circuit may not reset the microcontroller.

C. Component Tolerance

For this design, a 10 k Ω resistor with a tolerance of $\pm 1\%$ and a 220 nF capacitor with a tolerance of $\pm 10\%$ are used.

Since (13) can be written as

$$t_{\text{release}} = kRC, \quad (19)$$

where

$$k = -\ln\left(1 - \frac{V_{IH}}{V_0}\right), \quad (20)$$

the effect of changes in R and C can be written as

$$t + \Delta t = k(R + \Delta R)(C + \Delta C) \quad (21)$$

$$= kRC + kR\Delta C + kC\Delta R + k\Delta R\Delta C. \quad (22)$$

Ignoring the second-order term,

$$t + \Delta t \approx kRC + kR\Delta C + kC\Delta R. \quad (23)$$

Dividing by $t = kRC$ gives

$$\frac{\Delta t}{t} \approx \frac{\Delta R}{R} + \frac{\Delta C}{C}. \quad (24)$$

Using the component tolerances,

$$\frac{\Delta t}{t} \approx 0.01 + 0.10 = 0.11. \quad (25)$$

Therefore, the approximate timing tolerance is $\pm 11\%$.
Using the nominal release time of 2.65 ms,

$$t_{\min} \approx 2.65(1 - 0.11) \approx 2.36 \text{ ms}, \quad (26)$$

and

$$t_{\max} \approx 2.65(1 + 0.11) \approx 2.94 \text{ ms}. \quad (27)$$

Therefore, the reset release time is approximately between 2.36 ms and 2.94 ms.